Improved SAR Radiometric Cross-Calibration Method Based on Scene-Driven Incidence Angle Difference Correction and Weighted Regression

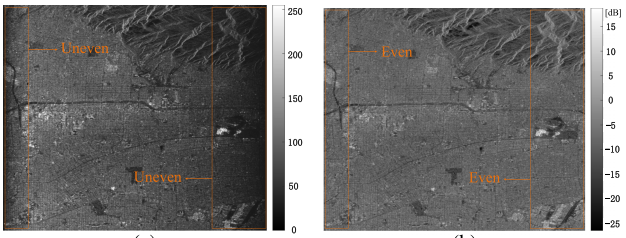
Yongsheng Zhou, *Member, IEEE*, Bopeng Yang, *Student Member, IEEE*, Qiang Yin, *Senior Member, IEEE*, Fei Ma, *Member, IEEE*, and Fan Zhang, *Senior Member, IEEE*

***Abstract*— Traditional absolute radiometric calibration methods for synthetic aperture radar (SAR) face challenges in terms of flexibility, maintenance, and calibration frequency. In contrast, radiometric cross calibration can achieve rapid and timely calibration by utilizing the calibrated SAR satellites to illuminate the same ground targets. However, there are still two factors limiting the accuracy of cross calibration. First, two satellites used for cross calibration often have different incidence angles, whereas the existing methods for correcting incidence angle differences have poor performance in scene adaptation and overcorrection. Second, the stability of ground targets plays a critical role in effective cross calibration, but in practice, not all targets possess the same stability. To address the first issue, this article proposes a novel scene-driven incidence angle difference correction method. It leverages the historical information about the target scenes to determine the evaluation threshold for data blocks. Moreover, it incorporates the adaptive exponential cosine model to correct the scattering variations caused by the difference in incidence angle. To address the second issue, an uncertainty analysis method is employed to calculate the uncertainty of each data block. Then, these uncertainties are utilized to calculate weight coefficients, and the calibration constant is determined using a weighted least squares (WLS) model. Cross-calibration experimental results on the Sentinel-1A/B demonstrate an average reduction of 21.6% in the relative calibration error and 18.6% in the root-mean-square error (RMSE) compared with the traditional method, validating the effectiveness of the proposed method.**

***Index Terms*— Incidence angle correction, radiometric cross calibration, synthetic aperture radar (SAR), weighted least squares (WLS) regression.**

I. Introduction

SYNTHETIC aperture radar (SAR) is an advanced remote sensing technology that employs radar systems to acquire high-resolution microwave images of the Earth's surface [1], [2]. Unlike optical remote sensing sensors, SAR operates effectively in all weather conditions and at any time of day, facilitating detailed observations of topography, vegetation, and artificial structures. It has been widely applied in various fields, such as environmental monitoring, disaster management, and military reconnaissance [3], [4], [5], [6], [7].



(a) (b)

Fig. 1. SAR images (a) before and (b) after radiometric calibration.

Radiometric calibration is essential for the processing and quantitative analysis of SAR data [8]. Through absolute radiometric calibration, digital numbers (DNs) in SAR imagery are converted into calibrated radiometric values, typically measured in physical units, such as radar cross section (RCS) or backscattering coefficient (expressed in dB, decibelized units), as shown in Fig. 1. Utilizing calibrated SAR data enables the generation of accurate and comparable radiometric information across various SAR images and datasets, which is crucial for conducting quantitative analyses and accurately monitoring and characterizing Earth's surface properties. Furthermore, calibrated SAR data also facilitate the intercomparison between different SAR sensors and missions, thereby contributing to the development of long-term and large-scale Earth observation datasets [9], [10], [11], [12], [13].

Traditional methods for SAR absolute radiometric calibration primarily rely on artificial calibrators, such as corner reflectors and active transponders, known for their well-documented reflectivity values [14], [15], [16], as shown in Fig. 2(a). However, the utilization of these artificial calibrators faces challenges when attempting to increase the calibration frequency [17]. The impracticality of incorporating a large number of calibrators for improving calibration frequency

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Yongsheng Zhou, Bopeng Yang, Qiang Yin, and Fei Ma are with the College of Information Science and Technology, Beijing University of Chemical Technology, Beijing 100029, China (e-mail: zhyosh@mail.buct.edu.cn; 2021210543@buct.edu.cn; yinq@mail.buct.edu.cn; mafei@mail.buct.edu.cn).

Fan Zhang is with the College of Information Science and Technology and the Interdisciplinary Research Center for Artificial Intelligence, Beijing University of Chemical Technology, Beijing 100029, China (e-mail: zhangf@mail.buct.edu.cn).

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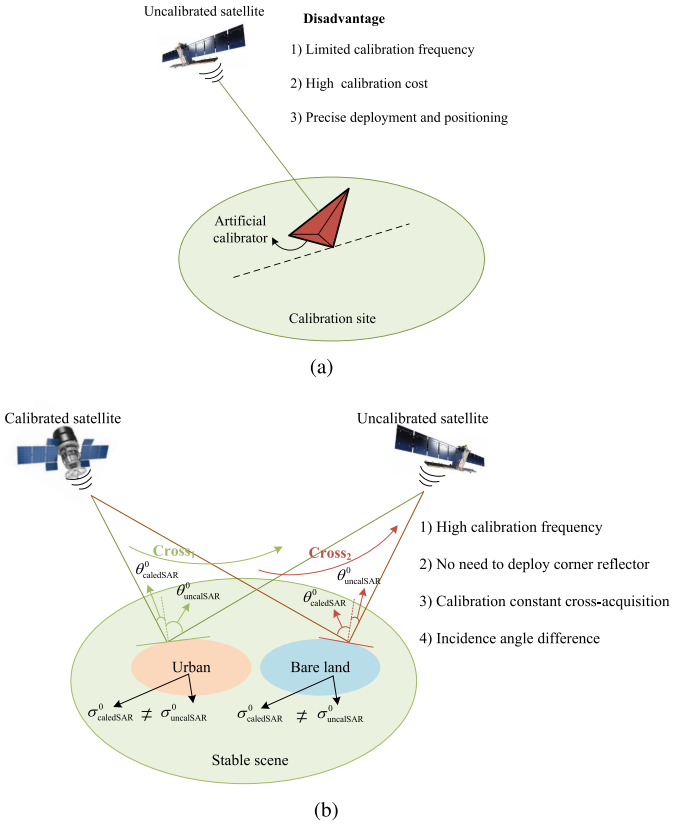


Fig. 2. Comparison between (a) traditional radiometric calibration method and (b) radiometric cross-calibration method.

is because of the following: 1) the high cost associated with manufacturing the calibrators; 2) the significant effort and resources required for their precise deployment and positioning; and 3) regular maintenance demands due to their susceptibility to disturbances or movements. Thus, there is a need to explore alternative calibration methods.

Radiometric cross calibration is an innovative approach that employs calibrated satellites to calibrate other satellites while simultaneously observing the same targets [18], [19]. Widely embraced in optical and meteorological satellites [20], [21], [22], this method has recently found application in SAR radiometric calibration [23], as shown in Fig. 2(b). Unlike traditional SAR radiometric calibration, the fundamental principle of radiometric cross calibration involves utilizing naturally distributed targets as target scenes, eliminating the need for deploying artificial calibrators and conducting field measurements. This approach reduces costs, requires no maintenance, and capitalizes on the abundance of naturally distributed targets for high-frequency calibrations. At the same time, since the cross scene is a uniformly distributed target, it can provide a certain degree of fault tolerance for the SAR image registration of the satellites on both sides of the cross calibration [24], [25]. The versatility of radiometric cross calibration positions it as one of the most promising calibration methods.

However, SAR radiometric cross calibration faces unique challenges due to the side-looking imaging characteristics of SAR and scattering inconsistencies among naturally distributed targets between calibrated and uncalibrated sensors. Specially, these challenges relate to two main aspects.

1) Differences in incidence angles between calibrated and uncalibrated SAR data can impact cross-calibration precision. During cross calibration, the target's true backscattering coefficients ($\sigma^0_{\text{uncalSAR}}$) under the uncalibrated SAR imaging geometry are derived from the calibrated SAR ($\sigma^0_{\text{caledSAR}}$), as shown in Fig 2(b). However, the incidence angles of the two SAR sensors are usually different, i.e., $\theta_{\text{uncalSAR}} \neq \theta_{\text{caledSAR}}$ [26], resulting in different backscattering characteristics of the same target ($\sigma^0_{\text{uncalSAR}} \neq \sigma^0_{\text{caledSAR}}$). Directly assuming $\sigma^0_{\text{uncalSAR}} = \sigma^0_{\text{caledSAR}}$ to cross calibrate the uncalibrated SAR would introduce errors. While several studies have examined the quantitative relationship between target backscatter intensity and incidence angle [27], [28], [29], these methods rely on specific SAR data and face challenges in different terrain scenes. Surface backscattering models were originally developed for ground-surface parameter inversion. The models express the relationship between the backscattering coefficient and the incidence angle. Therefore, Zhou et al. [23] used the Oh semiempirical model to derive an incidence angle difference correction method; however, its applicability is limited to specific terrain scenes, often resulting in overcorrection of specific data [30]. This study proposes a novel correction method that enhances adaptability across diverse terrains by employing a cosine model that evaluates scene characteristics to select model parameters adaptively.
2) The uncertainty associated with naturally distributed targets for cross calibration is generally greater than with traditional artificial calibrators. In traditional SAR radiometric calibration, DN-RCS data pairs from different artificial calibrators are employed with the ordinary least squares (OLSs) method to determine the calibration constant [31]. In SAR radiometric cross calibration, data pairs change to DN-$\sigma^0_{\text{uncalSAR}}$ data pairs of different distributed target blocks, with $\sigma^0_{\text{uncalSAR}}$ derived from calibrated SAR. However, the accuracy uncertainties of backscattering coefficients vary among blocks due to residual errors in incidence angle correction and other factors. Since OLS does not account for these varying uncertainties, calibration constant calculations may be affected. The weighted least squares (WLSs) are a variation of the OLS method used in regression analysis [32]. While OLS assumes that all DN-$\sigma^0_{\text{uncalSAR}}$ data pairs have equal importance in estimating the regression parameters, the use of WLS can assign different weights to different DN-$\sigma^0_{\text{uncalSAR}}$ data pairs based on their accuracy uncertainties. The determination of the weight is critical to the application of WLS. This study proposes a calculation method for the weight, incorporating uncertainties related to radiation stability, spatiotemporal consistency, and scene uniformity. By assigning appropriate weights and using WLS, we enhance the cross-calibration accuracy.

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In summary, this article presents the following contributions.

1) A scene-driven method for correcting incidence angle differences, addressing scattering variations caused by differences in incidence angles during cross calibration across multiple scenes. This method establishes an empirical relationship between the target area's DN and $\sigma^0_{\text{caledSAR}}$ by analyzing the historical SAR data of the data blocks. Subsequently, an evaluation threshold is determined to select data blocks that meet the incidence angle condition. Depending on the type of cross-scene features, the appropriate cosine power index n is determined to correct the incidence angle using the cosine model.
2) To address the accuracy disparities among distributed target blocks arising from variations in radiometric stability, temporal and spatial consistency, and scene uniformity, an uncertainty analysis method is utilized. This method calculates the associated uncertainty of the distributed target blocks and assigns weighting coefficients accordingly. Subsequently, the calibration constant of the uncalibrated SAR is determined using the WLS regression method.

The remainder of this article is structured as follows. Section II presents related work. Section III details the correction method for incidence angle difference between calibrated and uncalibrated SAR, and the calculation method of weight and corresponding WLS regression methods. Section IV presents the experimental results. Section V concludes this article.

II. Related Work

A. SAR Radiometric Cross Calibration

Traditional methods for SAR absolute radiometric calibration rely mainly on artificial calibrators, which limit their scalability for frequent calibrations. Radiometric cross calibration maximizes the utilization of information from previously calibrated SAR data. In the domain of SAR radiometric cross calibration, a seminal study was conducted by Zink et al. [33]. They compared data acquired by two distinct airborne SAR sensors at the exact location, thereby confirming the feasibility of effective cross calibration between disparate sensor images. The calibration accuracy could reach up to 2 dB due to variations in incidence angle, azimuth angle, and surface moisture. Wakabayashi et al. [34] conducted cross-calibration experiments in 2001 using an airborne L-band SAR system. By employing corner reflectors, they successfully calibrated SAR data and evaluated calibration accuracy by examining crosstalk and gain balance. In addition, in the same year, Uratsuka et al. [35] evaluated the AIRSAR and PI-SAR systems through cross validation, utilizing corner reflectors deployed within the scene. Alexander Zakharov and Sokolov [36] conducted a comparative study on the radiation stability of two SAR systems equipped with a parabolic antenna. However, this study relied on artificial calibration, which presented challenges compared with cross-calibration methods. Zhou et al. [23] investigated the radiometric cross calibration of the Sentinel series satellites using naturally stable distributed targets. The study successfully achieved cross calibration between two satellites, attaining a calibration accuracy within 1 dB. This highlights the growing importance of cross calibration with the increasing number of SAR satellites and the need for frequent calibration.

B. Correction of Incidence Angle Difference

The incidence angle is the angle between the radar beam and the normal (perpendicular) to the target surface. Targets exhibit varying backscattering characteristics at different incidence angles. Correcting for these differences ensures that the radiometric values in the image are consistent and accurately represent the backscattered signal strength. O'Grady et al. [37] developed a linear regression model to establish the relationship between the incidence angle θ and the backscattering coefficient σ^0. They utilized the coefficients obtained from the model as supplementary information to establish absolute thresholds, enabling the distinction of vegetation categories and geological features. Similarly, Widhalm et al. [38] utilized a linear approach to partially uncover the relationship between σ^0 and θ. However, this method overlooked the intricate physical mechanisms between the two variables. Moreover, it necessitated numerous auxiliary parameters, leading to high normalization accuracy but limited portability in specific experimental areas. Wu et al. [39] and Ye et al. [40] employed a histogram matching method to correct radar incidence angles. This method is easily applicable and requires minimal prior knowledge, as it is independent of sensor characteristics and ground conditions [27]. However, it is essential to note that the correction results may exhibit discontinuities in the range direction of SAR images [41]. Ulaby et al. [42] demonstrated that the relationship between σ^0 and θ follows the cosine law, derived from Lambert's law. This cosine-based model performs better for homogeneous pixels. However, Guo et al. [43] presented evidence suggesting that the power exponent of the cosine function is influenced by multiple factors, rendering it unsuitable to fulfill the requirements of different land cover types. The assumption of a cosine-dependent relationship does not hold well. Furthermore, using a fixed n value of the cosine model for SAR calibration is unsuitable for practical application due to the variability in land surface types, growth stages, and radar polarization modes. To address this issue, this study investigates the characteristics of various scenarios and establishes thresholds to filter distributed data blocks. It also incorporates an adaptive selection of the n value, thereby mitigating the problem of excessive data correction while simultaneously optimizing the correction effectiveness of the cosine function.

C. Calculation of Calibration Constant

The calibration constant is key to converting SAR data into physically meaningful backscattering values. It is typically derived from the slope of a linear regression between DN and radar backscatter coefficients. Vermote and Saleous [44] employed a multipoint linear regression method to perform cross calibration between NOAA-16/AVHRR and

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Terra/MODIS in 2006, using stable targets, such as deserts. To establish a cross-calibration system for the infrared band, Zhao et al. [45] utilized METOP IASI and AQUA AIRS as references in 2015. They calculated the calibration constant based on consistent spatial, temporal, and viewing angle observations. In the cross-calibration study on Beijing-1 small satellites in 2018, Chen et al. [46] utilized a geometric least squares method to calculate the calibration constant for the uncalibrated satellite by selecting well-matched data points. In the case of SAR satellites, Zhou et al. [23] developed a method for obtaining cross-calibration constant by performing absolute radiometric calibration. They employed natural stable distributed targets, calibration equations, and the least squares method to acquire the calibration constant for the uncalibrated satellite. It is worth noting that in practical applications, the accuracy of the calibration constant and the subsequent calibration performance can be influenced by factors, such as scattering stability, spatial matching, and site homogeneity. Therefore, it is crucial to analyze these influencing factors to ensure reliable calibration results. Using the uncertainty analysis method, this study quantifies the uncertainties associated with distributed target blocks resulting from radiometric stability, temporal and spatial consistency, and scene uniformity. Weighting coefficients are subsequently assigned based on these uncertainties. Subsequently, the calibration constant of the uncalibrated satellite is calculated using the WLS method.

III. Methodology

A. Overall Flowchart

This study focuses on reducing errors caused by variations in incidence angles between uncalibrated and calibrated SAR, while addressing uncertainty in quality across distributed target blocks for accurate calculation of radiometric calibration constant. Fig. 3 depicts the comprehensive flowchart of the improved SAR radiometric cross-calibration method. The process involves three key steps: the unique selection of distributed target blocks, the precise correction of the incidence angle difference driven by scene evaluation, and applying the WLS regression technique to calculate the calibration constant.

B. Step 1: Extraction of Data Blocks for Target Scene

In the process of SAR radiometric cross calibration, the uncalibrated SAR and the calibrated SAR both capture imagery of the same distribution target, which is usually the same land cover type. To ensure precise calibration of the uncalibrated SAR, the backscattering characteristics of the distribution target must remain stable [23], [47], [48]. Chen et al. [49] investigated and discussed the temporal stability of scattering for different land covers using Sentinel-1 data. They identified multiple natural targets suitable for cross calibration.

Due to the speckle noise problem, for a uniformly distributed target, its backscattering properties are usually expressed using statistical values rather than independent pixel values. Analogous to conventional calibration techniques relying on artificial calibrators, which employ various calculation methods, such as integration and peak value methods for point targets, diverse backscattering property calculation methods are available for distributed targets. The selection of the calculation method can exert an influence on the ultimate calibration accuracy.

This study investigates the impact of different data block extraction methods on the temporal backscattering characteristics of homogeneous areas within the target scene. The goal is to identify the optimal method for extracting distribution data blocks for cross calibration.

First, a set of SAR image data is acquired that shares the same ascending or descending orbit, incidence angle, imaging area, and polarization mode. These images are captured at an approximate interval of three months. Subsequently, the image dataset undergoes preprocessing steps, including orbit correction, thermal noise removal, radiometric calibration, speckle filtering, and geocoding. Next, homogeneous scenes within the images are segmented, and the median and mean values of the data blocks are calculated. The stability of the backscattering coefficient values obtained from these two methods is compared over the time series. A more minor error in the time series of the backscattering coefficient indicates higher temporal stability. Consequently, the corresponding values are considered descriptive characteristics of the backscattering coefficient for the target scene. The error is calculated using the standard deviation (STD)

$$\mathrm{STD} = \sqrt{\frac{1}{N} \cdot \sum_{i=1}^{N} \left(\sigma_i^0 - \sigma_{\mathrm{AVG}}^0\right)^2} \tag{1}$$

where σ_i^0 is the backscattering coefficient of the ith data block and σ_{AVG}^0 is the average backscattering coefficient value of all data blocks. N is the total number of data blocks.

The descriptive features of suitable backscattering coefficients in different scenarios, derived from the above analysis, are utilized to extract distributed data blocks from the target scene. Collecting calibrated and uncalibrated SAR data within a short period is crucial to mitigating the potential effects of terrain changes and rainfall fluctuations. The target scene region of this article contains target subregions with higher and lower backscattering coefficients, i.e., the urban scene and bare land scene regions used subsequently. The data blocks within the target scene region are divided as follows: 20 groups of data blocks are delineated within the homogeneous region of each of the above two types of target subregions to obtain enough samples for the fitting of the calibration constants. The size of each data block is 20 × 20 pixels. The reason for this division is to ensure that each group of data blocks has different scattering characteristics in the corresponding scene in order to better fit the exact calibration constants.

C. Step 2: Scene-Driven Incidence Angle Difference Correction

While the latest solution [23], relying on the Oh semiempirical model, has achieved the correction accuracy of the incidence angle difference to a certain degree, its

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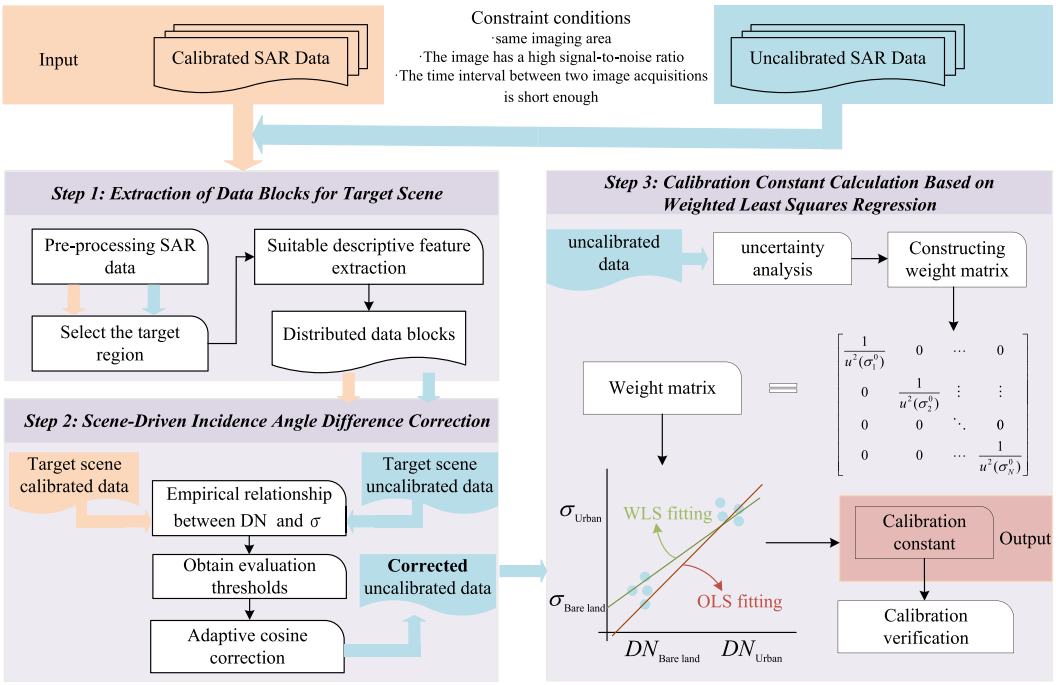


Fig. 3. Overall flowchart of the proposed method.

applicability is limited to specific ground target scenes. Moreover, it overlooks the potential overcorrection resulting from distributed data blocks [30]. This article introduces an effective method to mitigate the impact of incidence angle differences on SAR radiometric cross calibration through scene evaluation. The scenes associated with these data blocks undergo an evaluation to determine the evaluation threshold. Finally, the correction of the incidence angle difference is executed by employing an adaptive cosine model.

1) Determining the Threshold for Scene Evaluation Based on Distributed Data Blocks: The existing incidence angle correction approach [23] involves correcting all cross-calibration data blocks, regardless of their sensitivity to variations in the incidence angle and the minimal differences in their backscattering coefficients. In Fig. 4(a), two sets of data blocks, enclosed within the orange dashed elliptical box, are observed to be closely aligned with the central axis. The proximity to the central axis suggests that the backscattering coefficients of these data blocks exhibit limited sensitivity to changes in the incidence angle. Applying the incidence angle correction model directly to these data blocks may move them away from the central axis, leading to overcorrection and potentially compromising calibration accuracy.

This article tackles the aforementioned issue by integrating empirical relationships derived from historical data with a threshold method to identify the data blocks requiring incidence angle correction.

In the initial stage, historical image data from both calibrated and uncalibrated SAR are obtained, each having distinct incidence angles for the same target scene.

Subsequently, the amplitude values (namely, DN values) and backscattering coefficients are extracted from the urban area and bare land area, respectively.

Following this, the disparities in backscattering coefficients and amplitudes are computed for various land cover types within the same target scene and expressed in decibels (dB):

$$\Delta\sigma^{0_{\text{Bareland}}} = \sigma^{0_{\text{Bareland}}}_{\text{caledSAR}} - \sigma^{0_{\text{Bareland}}}_{\text{uncalSAR}} \tag{2}$$

$$\Delta\sigma^{0_{\text{Urban}}} = \sigma^{0_{\text{Urban}}}_{\text{caledSAR}} - \sigma^{0_{\text{Urban}}}_{\text{uncalSAR}} \tag{3}$$

$$\Delta\text{DN}^{\text{Bareland}} = \text{DN}^{\text{Bareland}}_{\text{caledSAR}} - \text{DN}^{\text{Bareland}}_{\text{uncalSAR}} \tag{4}$$

$$\Delta\text{DN}^{\text{Urban}} = \text{DN}^{\text{Urban}}_{\text{caledSAR}} - \text{DN}^{\text{Urban}}_{\text{uncalSAR}} \tag{5}$$

where $\sigma^{0_{\text{Bareland}}}_{\text{caledSAR}}$ and $\sigma^{0_{\text{Bareland}}}_{\text{uncalSAR}}$ represent the backscattering coefficients obtained from the calibrated and uncalibrated SAR, respectively, in the bare land area. Similarly, $\sigma^{0_{\text{Urban}}}_{\text{caledSAR}}$ and $\sigma^{0_{\text{Urban}}}_{\text{uncalSAR}}$ represent the backscattering coefficients obtained from the calibrated and uncalibrated SAR, respectively, in the urban area. In addition, $\text{DN}^{\text{Bareland}}_{\text{caledSAR}}$ and $\text{DN}^{\text{Bareland}}_{\text{uncalSAR}}$ correspond to the amplitude values acquired from the calibrated and uncalibrated SAR, respectively, in the bare land area. Likewise, $\text{DN}^{\text{Urban}}_{\text{caledSAR}}$ and $\text{DN}^{\text{Urban}}_{\text{uncalSAR}}$ indicate the amplitude values obtained from the calibrated and uncalibrated SAR, respectively, in the urban area. Moreover, $\Delta\sigma^{0_{\text{Bareland}}}$ and $\Delta\sigma^{0_{\text{Urban}}}$ represent the differences in backscattering coefficients between the calibrated and uncalibrated SAR in the corresponding bare land and urban area. Similarly, $\Delta\text{DN}^{\text{Bareland}}$ and $\Delta\text{DN}^{\text{Urban}}$, respectively, represent the differences in amplitude values between the calibrated satellite and the uncalibrated satellite in the corresponding bare land and urban area.

To assess the variations in changes pre- and post-correction of backscattering coefficients and amplitudes, it is possible to analyze the empirical relationships regarding their disparities. Specifically, linear fits can be derived for the distinct scenarios of bare land and urban areas

$$\Delta\text{DN}^{\text{Bareland}} = a_{\text{Bareland}} \cdot \Delta\sigma^{0_{\text{Bareland}}} + b_{\text{Bareland}} \tag{6}$$

$$\Delta\text{DN}^{\text{Urban}} = a_{\text{Urban}} \cdot \Delta\sigma^{0_{\text{Urban}}} + b_{\text{Urban}} \tag{7}$$

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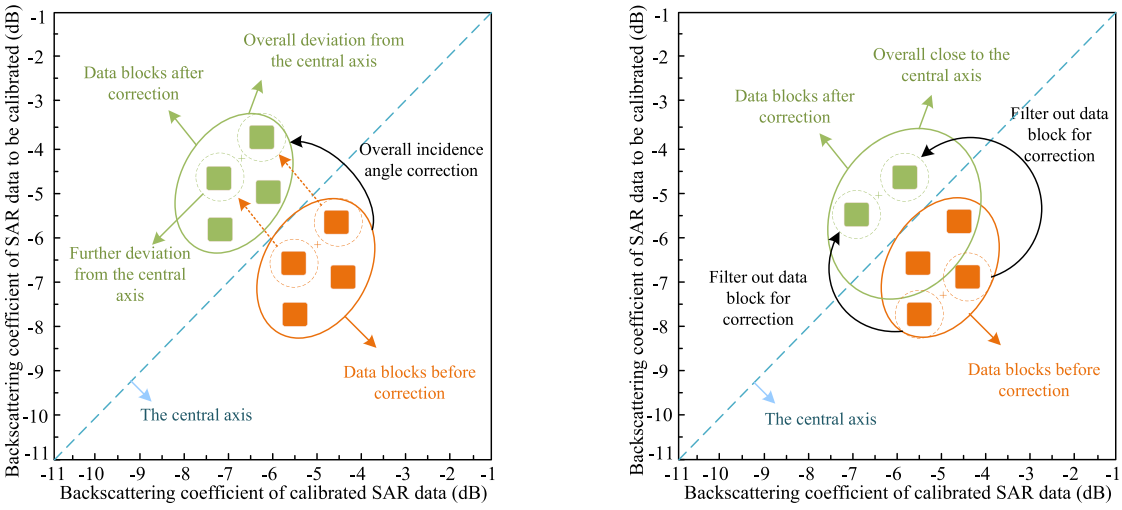


Fig. 4. Different methods for incidence angle correction of distributed data blocks. (a) Traditional approach. (b) Scene-driven approach.

where a and b represent the slope and intercept of the linear regression between ΔDN and $\Delta\sigma^0$.

The analysis of historical data from the target scene shows that when the differences in backscattering coefficients caused by variations in the incidence angle are within 0.3 dB, the relative error before and after calibration is minimized. Therefore, the correction for incidence angle difference is not necessary. As depicted in Fig. 4(b), it can be observed that within the orange solid line elliptical box, only the data blocks enclosed by the orange dashed line box undergo correction. This selection is based on the fact that these two sets of data blocks exhibit a more significant deviation from the central axis and are consequently filtered out for correction. After the correction of all data blocks, it is observed that they exhibit more excellent proximity to the central axis compared with the overall corrected data blocks depicted in Fig. 4(a). This article sets a threshold of 0.3 dB for the differences in backscattering coefficients. To ensure the selection of appropriate data, the differences in amplitude values before and after calibration are evaluated for each data block. For bare land, data with amplitude value differences exceeding $(0.3 \cdot a_{\text{Bareland}} + b_{\text{Bareland}})$ at different incidence angles require correction, while those below the threshold do not. Similarly, for urban area, data with amplitude value differences exceeding $(0.3 \cdot a_{\text{Urban}} + b_{\text{Urban}})$ at different incidence angles require correction, while those below the threshold do not.

2) Correcting Scattering Differences at Different Incidence Angles: One of the earliest and most widely adopted methods for incidence angle correction in SAR is the cosine correction method [42], derived from Lambert's law in optics. This method is based on the fundamental assumption that the power energy of the incident radiation obeys the cosine law when it is reilluminated on the upper hemisphere. Moreover, considering that the rate of radiation change is also cosine-dependent on the observed area, the backscattering coefficients of both calibrated and uncalibrated SAR images can be expressed as follows:

$$\sigma^0_{\text{caledSAR}} = \sigma^0_{\text{indep}} \cdot \cos^n(\theta_{\text{caledSAR}}) \tag{8}$$

$$\sigma^0_{\text{uncalSAR}} = \sigma^0_{\text{indep}} \cdot \cos^n(\theta_{\text{uncalSAR}}) \tag{9}$$

where σ^0_{indep} represents the backscattering coefficient that is independent of the incidence angle θ within the same target scene. n represents the adjustment factor for the cosine function, which accounts for the dependence of the backscattering coefficient σ^0 on the incidence angle θ. The backscattering coefficients of calibrated SAR data and uncalibrated SAR data are denoted as $\sigma^0_{\text{caledSAR}}$ and $\sigma^0_{\text{uncalSAR}}$, respectively. Furthermore, the radar incidence angle for the calibrated SAR data is represented by θ_{caledSAR}, while θ_{uncalSAR} corresponds to the radar incidence angle for the uncalibrated SAR data.

Based on (8) and (9), it can be inferred that the backscattering coefficients $\sigma^0_{\text{caledSAR}}$ acquired at an incidence angle of calibrated SAR can be effectively converted to the backscattering coefficients $\sigma^0_{\text{uncalSAR}}$ at any incidence angles of an uncalibrated SAR. The conversion expression is

$$\sigma^0_{\text{uncalSAR}} = \frac{\cos^n(\theta_{\text{uncalSAR}})}{\cos^n(\theta_{\text{caledSAR}})} \cdot \sigma^0_{\text{caledSAR}}. \tag{10}$$

Surface roughness significantly impacts the performance of the cosine correction method, rendering a single model inadequate for universal application across all land cover scenarios depicted in the image. The primary objective of this article is to address the scattering changes induced by incidence angles in both bare land and urban area, with a particular focus on the effectiveness of the adaptive exponential cosine model technique. An analysis of historical data, as shown in Fig. 5, reveals that the cosine correction method yields a minor incidence angle correction error for bare land when the parameter n is set to 5. Similarly, for urban area, the method exhibits the most minor correction error when the parameter n is set to 2.

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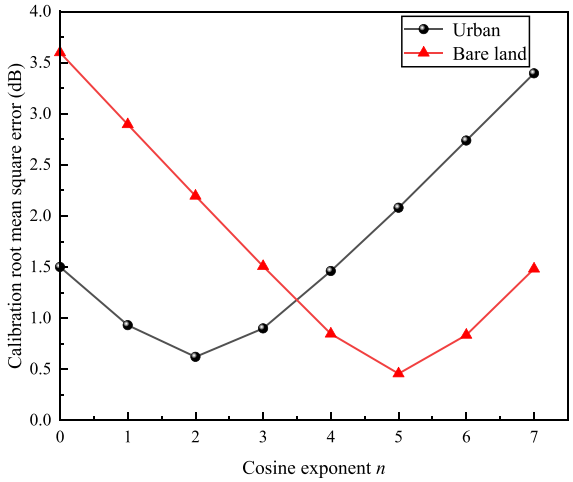


Fig. 5. Incidence angle correction RMSEs of urban and bare land under different n values.

The detailed correction process of scattering differences at different incidence angles is shown in Algorithm 1.

Algorithm 1 Alleviating the Impact of Incidence Angle Differences Through Scene-Based Assessment

Input: Backscattering coefficients of historical SAR image datasets σ_1 σ_2, amplitude value DN_1 DN_2; Amplitude values of calibrated SAR data $DN_{caledSAR}$ / incidence angle $\theta_{caledSAR}$; Incidence angle of uncalibrated SAR data $\theta_{uncalSAR}$; Cross-calibration data blocks ($\sigma_{caledSAR}$, $DN_{uncalSAR}$); Target scene T_0;

Output: Cross-calibration data blocks after incidence angle difference correction ($\sigma_{correct}$, $DN_{uncalSAR}$);

if $T_0 = T_{\text{Bare land}}$ **then**
  $|\sigma_1 - \sigma_2|$, $|DN_1 - DN_2| \xrightarrow{f}$ $ThresholdValue_1$;
  if $|DN_{caledSAR} - DN_{uncalSAR}| > ThresholdValue_1$ **then**
    $\sigma_{correct} = [\cos(\sigma_{uncalSAR})/\cos(\sigma_{caledSAR})]^n \cdot \sigma_{caledSAR}$, $n = 5$;
  else
    $\sigma_{correct} = \sigma_{caledSAR}$;
  end if
else
  $//T_0 = T_{\text{Urban}}$
  $|\sigma_1 - \sigma_2|$, $|DN_1 - DN_2| \xrightarrow{f}$ $ThresholdValue_2$;
  if $|DN_{caledSAR} - DN_{uncalSAR}| > ThresholdValue_2$ **then**
    $\sigma_{correct} = [\cos(\sigma_{uncalSAR})/\cos(\sigma_{caledSAR})]^n \cdot \sigma_{caledSAR}$, $n = 2$;
  else
    $\sigma_{correct} = \sigma_{caledSAR}$;
  end if
end if

D. Step 3: Calibration Constant Calculation Based on WLS Regression

The traditional calibration or cross-calibration methods commonly utilize the OLS regression approach to calculate the calibration constants. These methods are mainly based on calibration data (different distributed data blocks) that are closely matched in terms of time, space, and observation geometry. However, these methods overlook the inherent variations in quality uncertainty among the data blocks, thereby diminishing the accuracy of the resulting calibration constant.

To tackle this limitation, a novel approach is proposed in this article, which fully utilizes the advantages of WLS in accounting for heteroscedasticity, handling outliers, and flexibility in specifying weights [32]. Weight coefficients are assigned to each cross-calibration data block. These coefficients are determined by considering the uncertainties associated with the radiometric stability, temporal and spatial consistency, and scene uniformity of the respective data blocks.

1) Radiometric Cross-Calibration Equation: The primary objective of SAR radiometric calibration is to establish the relationship between the target's amplitude value (DN) and its radar scattering cross section (σ). By the radar equation [50], [51], the image pixel power can be expressed as follows:

$$P = \frac{P_t g_r G_r(\theta,\varphi) g_t G_t(\theta,\varphi) \lambda^2 k}{(4\pi)^3 R^4} \cdot \sigma \tag{11}$$

where P represents the image power, P_t represents the transmit power, g_r characterizes the receiving antenna gain, g_t characterizes the transmitting antenna gain, G_r characterizes the receiving antenna gain pattern, G_t characterizes the transmitting antenna gain pattern, λ represents the radar wavelength, R is the slant range, σ represents the target's RCS, θ represents the incidence angle, φ represents the azimuth angle, and k is the proportional coefficient that converts the received power into image power.

In general, the receiving and transmitting antennas are typically the same. Taking into account system losses (L_s) and radio wave attenuation (L_α), (11) can be expressed as follows:

$$P = \frac{P_t g^2 G^2 \lambda^2 k}{(4\pi)^3 R^4 L_s L_\alpha} \cdot \sigma. \tag{12}$$

By establishing the correlation between P and σ, one can derive the overall transfer function of the SAR system as follows:

$$P = H \cdot \sigma \tag{13}$$

where H represents the system transfer function

$$H = \frac{P_t g^2 G^2 \lambda^2 k}{(4\pi)^3 R^4 L_s L_\alpha}. \tag{14}$$

The varying parameters within the transfer function necessitate measurement or correction. Specifically, the system loss (L_s), radio wave attenuation (L_α), and antenna gain (g) are determined through internal calibration [53], while the antenna gain pattern is ascertained via relative radiometric calibration [54], [55].

In the SAR image, the square of any pixel's DN is directly proportional to the signal power it receives. The expression for this relationship can be written as follows:

$$DN^2 = K \cdot P \tag{15}$$

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where K represents the coefficient of proportionality between the square of the DN value and the signal power.

Combining (13)–(15), the relationship between the DN^2 amplitude value and its radar scattering cross section σ can be written as follows:

$$DN^2 = \frac{P_t g^2 G^2 \lambda^2 k}{(4\pi)^3 R^4 L_s L_\alpha} \cdot K \cdot \sigma \tag{16}$$

where $\sigma = \sigma^0 \cdot A$ and A represents the area occupied by the target scene region. Therefore, it can be obtained

$$DN^2 = \frac{P_t g^2 G^2 \lambda^2 k}{(4\pi)^3 R^4 L_s L_\alpha} \cdot A \cdot K \cdot \sigma^0. \tag{17}$$

From (17), it can be seen that there is a certain linear relationship between DN and σ^0, and the variable a can be used as the linear coefficient between them. Combining the idea of crossover, the relationship between the value of the uncalibrated target image DN_{uncal} and its backscattering coefficients $\sigma^0_{correct}$ can be obtained as follows:

$$\sigma^0_{correct} = a \cdot DN^2_{uncal}. \tag{18}$$

In an ideal scenario, the line passes through the origin (0,0). However, when considering errors, it is necessary to add an intercept term, denoted as b. The calibration equation is then represented as follows:

$$\sigma^0_{correct} = a \cdot DN^2_{uncal} + b \tag{19}$$

where $\sigma^0_{correct}$ represents the value of backscattering coefficients of the uncalibrated SAR data, calculated by the correction in Section III-B, and is equivalent to $\sigma^0_{uncalSAR}$. DN^2_{uncal} refers to the digital quantized value of the corresponding data block in the uncalibrated SAR data. a represents the calibration constant of the uncalibrated SAR data, while b represents the calibration offset of the uncalibrated SAR data.

When fitting a straight line, two points are needed to determine the line's slope and y-intercept. These two points provide the necessary information to uniquely identify the line. Ideally, these two points should have a large distance between them. This is because a larger distance between the points increases the precision and accuracy of the estimated line. This article determines the calibration constant by utilizing data acquired from urban area and bare land. Urban area data, characterized by high backscattering coefficients, and bare land data, characterized by low backscattering coefficients, are utilized to establish the calibration constant curve. The data consisting of 20 data blocks from urban area and 20 data blocks from bare land scenes are utilized for establishing calibration equation. The specific implementation process is visually depicted in Fig. 3.

2) Uncertainty Calculation for Cross-Calibration Data Blocks: The quality uncertainty of the data blocks mainly arises from the instability of target scattering, errors caused by spatial position matching, and a lack of uniformity in the data blocks. Assuming that the above errors are independent, the total uncertainty can be calculated using the square root of the sum of squares

$$u(\sigma^0) = \sqrt{u^2(\sigma^0_1) + u^2(\sigma^0_2) + u^2(\sigma^0_3)} \tag{20}$$

where $u(\sigma^0_x)(x = 1, 2, 3)$ represents the uncertainty of the three factors mentioned above.

1) $u(\sigma^0_1)$—*Scattering Stability:* The fluctuation of the backscattering coefficient within the scene over time affects the data block quality, leading to calibration errors. The main purpose of the scattering stability analysis of the data blocks compared with the scattering stability analysis of the target scene in Section III-C is to quantify the uncertainty brought by the scattering stability to the data blocks, while the main purpose of the latter is to analyze the stability of each data extraction method on the time series under different scenarios, to determine the data blocks extraction method that is compatible with the target scene.
2) $u(\sigma^0_2)$—*Spatial Matching:* Geometric correction and other factors may introduce pixel deviations in SAR imagery, highlighting the importance of selecting a site with homogeneous scattering properties for calibration. The geographic location of the SAR scene under calibration is adopted as the reference value to address this issue. These discrepancies can be transformed into actual distance deviations by analyzing the differences in pixel latitude and longitude between the calibrated SAR image and the uncalibrated SAR image in the same scene. Consequently, the relative distance deviation between the real-world geographic spaces of the uncalibrated SAR scene and the calibrated SAR scene indicates the uncertainty associated with spatial matching.
3) $u(\sigma^0_3)$—*Data Block Uniformity:* Analyzing cross data can provide valuable insights into the spatial variation of surface scattering characteristics within the corresponding scene, enabling a comprehensive assessment of the spatial heterogeneity of the target area. To quantify the uncertainty associated with data homogeneity, the coefficient of variation (CV), defined as the ratio of the STD to the mean of the original scattering data, is employed

$$CV = \frac{STD_{ts}}{\sigma^0_{Mean}} \tag{21}$$

where STD_{ts} represents the STD of the backscattering coefficient of the data block over the time series and σ^0_{Mean} represents the mean backscattering coefficient of the data block. The spatial uniformity of the target area improves as the CV decreases.

3) Establishment of Weight Coefficient Matrix for Data Blocks: The weight coefficients corresponding to cross-calibration data blocks are generally represented by the reciprocal of the square of their uncertainties. If the uncertainty of the nth data block is denoted as $u_n(\sigma^0)$, then the corresponding weight coefficient is represented as $1/u^2_n(\sigma^0)$.

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With a total of 40 data blocks, the weights for each can be represented using the weight coefficient matrix, denoted as W

$$w = \begin{bmatrix} \frac{1}{u_1^2(\sigma^0)} & 0 & \cdots & 0 \\ 0 & \frac{1}{u_2^2(\sigma^0)} & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \frac{1}{u_{40}^2(\sigma^0)} \end{bmatrix} \tag{22}$$

4) Calculation of the Calibration Constant of the Uncalibrated SAR: WLS [32] is adopted to calculate the calibration constant by assigning suitable weights to each set of data blocks. Unlike OLS, WLS enhances the accuracy of regression outcomes by utilizing weight coefficients. The weight coefficients for the data block are commonly determined by taking the reciprocal of the squared uncertainties attributed to various factors. Hence, when the uncertainties in the calibration steps of the data block are more significant, suggesting inferior quality, the corresponding weight coefficients become smaller. As a result, the impact of these data blocks on the calibration constant's regression is diminished.

This article utilizes data comprising urban and bare land scenes to establish the calibration constant for the uncalibrated SAR data, employing the calibrated SAR data. The urban scenes yield high backscattering coefficients, whereas the bare land scenes exhibit low backscattering coefficients. After scene evaluation, the selected data block $(\sigma^0_{\text{correct}}, \text{DN}_{\text{uncal}})$, denoted as $\sigma^0_{\text{correct}}$ for the calibrated backscattering coefficient corrected for incidence angle and DN_{uncal} for the amplitude value of the uncalibrated SAR data, is subjected to a WLS model. Through the minimization of the sum of squared generalized residuals, the model coefficients can be determined

$$\alpha = ((\text{DN}^2_{\text{uncal}})^{\text{T}} W (\text{DN}^2_{\text{uncal}}))^{-1} (\text{DN}^2_{\text{uncal}})^{\text{T}} W \cdot \sigma^0_{\text{correct}} \tag{23}$$

IV. Experiment and Discussion

In this section, experiments are conducted utilizing the Sentinel-1A/B dataset to quantitatively assess the effectiveness and accuracy of the proposed method. The temporal stability of urban area and bare land is investigated using various extraction methods. The calibration accuracy of different incidence angle correction methods is also evaluated. The calibration constant is calculated by employing the WLS regression to account for the uncertainty of the combined distributed data blocks. Subsequently, the calibration of Sentinel-1B data is carried out using Sentinel-1A data in the Said Port area of Egypt, and the accuracy of the calibration constant is validated.

A. Experimental Area and Data

1) Experimental Area: The experimental area for this study is situated in the Said Port region of Egypt, which encompasses a diverse set of land cover types, including urban area, bare land, deserts, water bodies, and farmland, as shown in Fig. 6. Since the background noise usually affects the accuracy of the image response energy extraction, urban area and bare land with relatively high backscattering coefficient are selected as the experimental region. Urban area exhibits a well-organized arrangement of buildings, while the bare land showcases a relatively flat topography with an average annual precipitation of 150 mm. These distinct characteristics make urban area and bare land particularly suitable for SAR cross calibration.

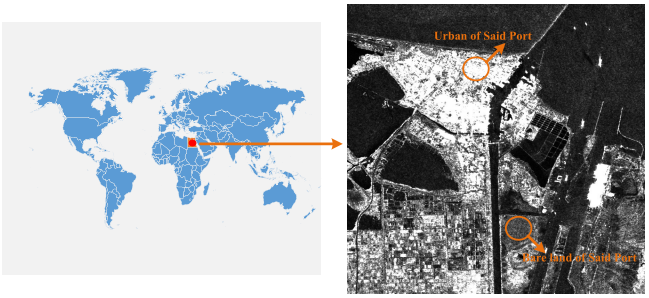


Fig. 6. SAR image of the experimental area in this article.

TABLE I
Parameters of Sentinel-1 IW GRD Data

Parameter	Value
Radar band	C band
Polarization mode	VV+VH
Center frequency	5.405GHz
Spatial resolution	10m
Revisit period	12 days for a single satellite 6 days for a double satellite
Swath	250km

2) Experimental Data: The Sentinel-1 (S-1) mission, jointly undertaken by the European Space Agency and the European Commission, comprises two satellites, Sentinel-1A, launched in April 2014, and Sentinel-1B, launched in April 2016. Each satellite operates at a frequency of 5.405 GHz and guarantees a revisit time of 12 days, ensuring frequent and consistent imaging of the Earth's surface. The Sentinel-1 satellites encompass a range of data acquisition modes, namely, stripmap mode (SM), extra-wide swath (EW) mode, interferometric wide swath mode (IW), and wave mode (WM). In particular, the IW mode finds prominent usage in land observations. This study specifically concentrates on the analysis of ground range detected (GRD) products acquired through the IW mode of the Sentinel-1 satellite, as evidenced by the details provided in Table I.

The experimental area is comprehensively covered by the Sentinel-1 IW GRD data sourced from ASF. Since the experimental area selected in this article is relatively homogeneous and has moderate surface roughness, under this condition, the electromagnetic waves in VV polarization mode show more significant vertical components, which can bring more abundant backscattered echoes. Thus, this study utilizes VV-polarized Sentinel-1 data to minimize background noise interference.

To ensure data accuracy and reliability, preprocessing steps, including orbit correction, thermal noise removal, radiometric calibration, and terrain correction, are performed using the Sentinel-1 toolbox.

B. Error Evaluation Indicators

The accuracy of the calibration constant obtained through the WLS method is verified by assessing two indicators: the relative error (ε) and the root-mean-square error (RMSE). ε indicates the proportion of the absolute error in the

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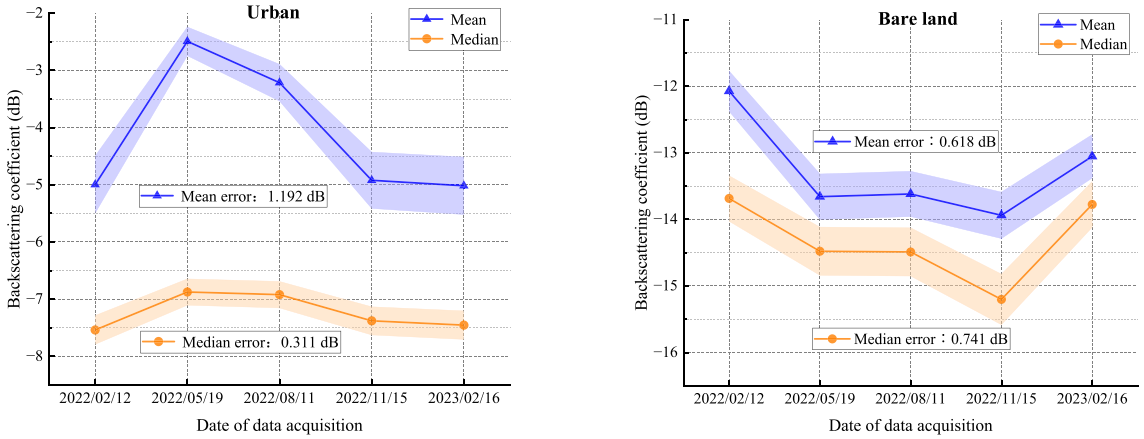


Fig. 7. Error analysis under different extraction methods of backscattering coefficient. (a) Time series in the urban. (b) Time series in the bare land.

measurement to the actual value of the measurement. On the other hand, RMSE measures the disparity between the calculated and actual values. A smaller value indicates a lesser difference between the two. A smaller error value indicates higher accuracy in the calculated calibration constant. The calculation formulas of ε and RMSE are as follows:

$$\varepsilon = \frac{1}{N} \cdot \sum_{i=1}^{N} \frac{\left|\sigma^0_{\text{act}_i} - \sigma^0_{\text{com}_i}\right|}{\sigma^0_{\text{act}_i}} \tag{24}$$

$$\text{RMSE} = \sqrt{\frac{1}{N} \cdot \sum_{i=1}^{N} \left(\sigma^0_{\text{act}_i} - \sigma^0_{\text{com}_i}\right)^2} \tag{25}$$

where $\sigma^0_{\text{act}_i}$ represents the actual backscattering coefficient corresponding to the ith pixel of the verification scene, $\sigma^0_{\text{com}_i}$ represents the backscattering coefficient corresponding to the ith pixel of the verification scene computed by the calibration constant, and N represents the total number of pixels occupied by the verification scene.

C. Extraction Results of Distributed Data Blocks

1) Determine the Appropriate Target Scene Extraction Method: In this section, we experiment to evaluate the impact of the mean and median methods on the temporal backscattering stability of urban area and bare land scenes. To ensure consistency, we utilize five sets of Sentinel-1A images with the same scene and incidence angle. Detailed information regarding the dataset is listed in Table II. For more precise registration, the data block sizes are initially cropped differently. After the registration is completed, all data blocks that are used as part of the scattering stability analysis are uniformly resized to 300 × 300 pixels, a size sufficient to cover the target scene area. If a region below this size is selected for analysis, the scattering stability analysis of the scene may be affected by the number of samples, noise effects, and edge effects, which may not correctly reflect the stability of the scene. Therefore, the entire 300 × 300 pixels window is used in this article to evaluate the temporal scattering stability of the target scene region.

TABLE II
SPECIFIC INFORMATION ABOUT THE SENTINEL-1A DATASET

Scene type	Obtaining time	Polarization	Data size
Urban+Bare land	2022/02/12	VV	300×300
Urban+Bare land	2022/05/19	VV	300×300
Urban+Bare land	2022/08/11	VV	300×300
Urban+Bare land	2022/11/15	VV	300×300
Urban+Bare land	2023/02/16	VV	300×300

Fig. 7 shows the extracted backscattering coefficients at different scenes. According to Fig. 7(a), the error in urban areas using the median method is 0.881 dB lower than that of the mean method. Therefore, for urban area scenes, the median method is preferred for extracting data blocks. As depicted in Fig. 7(b), in the bare land region, the error obtained by the mean method is only 0.123 dB lower than that obtained by the median method, while the mean method can provide data blocks with higher backscattering coefficients and reduce the influence of background noise, which can improve the accuracy of the subsequent experiments. Based on this observation, the mean method is selected for extracting data blocks in bare land scenes.

2) Extract Distributed Cross Data Blocks: Using appropriate extraction methods, data blocks are extracted for urban area and bare land. Two sets of image data, namely, the calibrated and uncalibrated data, are obtained from the same target scene for cross calibration. Since the target scene occupies a relatively small area within the entire image, the variation of incidence angles along range direction for this region is small, confined to just 0.5°. Consequently, the mean incidence angle within this range is adopted as the representative incidence angle for the region. Data are collected during similar period to reduce the effects of topographic changes and rainfall variations. Table III shows that the calibrated SAR has an incidence angle of about 33.1°, and the uncalibrated one is around 44.6°. The difference in incidence angles, approximately 11.5°, indicates the necessary correction magnitude for the incidence angle difference [23].

Landmark features are used for initial rough registration to locate target scenes in datasets, followed by slicing. Precise

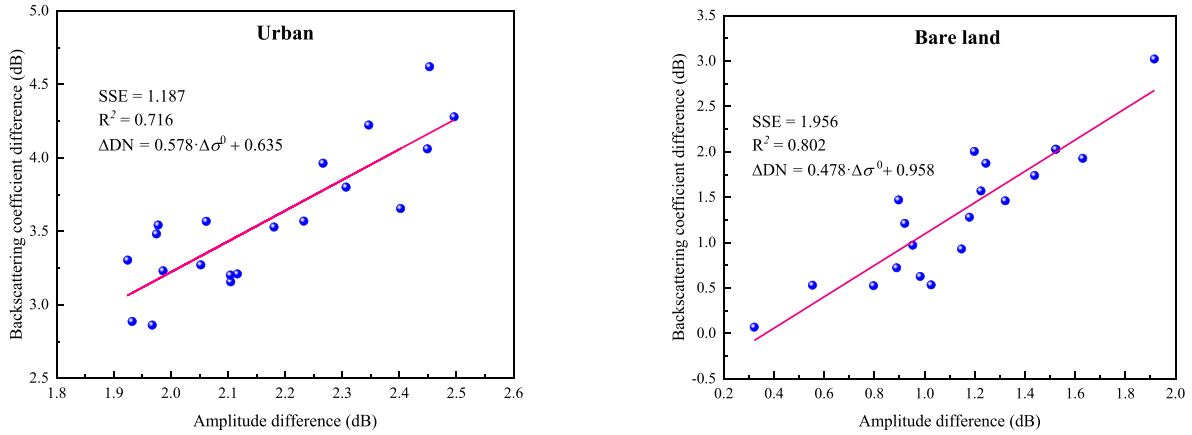
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TABLE III
SPECIFIC INFORMATION ABOUT THE DATASET USED TO EXTRACT THE DATA BLOCKS

Data	Scene type	Polarization mode	Obtaining time	Incidence angle	Satellite
Data 1	Urban+Bare land	VV	2021/08/15	44.6°	Sentinel-1B
Data 2	Urban+Bare land	VV	2021/08/16	33.1°	Sentinel-1A

TABLE IV
SPECIFIC INFORMATION ABOUT THE INCIDENCE ANGLE CORRECTION DATASET

Scene type	Satellite	Obtaining time	Polarization mode	Flight direction	Incidence angle
Urban+Bare land	Sentinel-1A	2021/02/12	VV	Ascending	33°
Urban+Bare land	Sentinel-1A	2021/05/19	VV	Ascending	33°
Urban+Bare land	Sentinel-1B	2021/02/16	VV	Descending	33.3°
Urban+Bare land	Sentinel-1B	2021/05/08	VV	Descending	33.3°



(a) (b)

Fig. 8. Empirical relationship between amplitude difference and backscattering coefficient difference. (a) Empirical relationship in the urban scene. (b) Empirical relationship in the bare land scene.

registration is done using the maximum correlation coefficient method, which the following formula can express:

$$R(u', v') = \frac{\sum_i \sum_j M_1(i, j) M_2(i + u', j + v')}{\sqrt{\sum_{ij} M_2(i, j)} \cdot \sqrt{\sum_{ij} M_2(i + u', j + v')}} \quad (26)$$

where u' represents the offset in the range direction and v' represents the offset in the azimuth direction. The DN values of the calibrated and uncalibrated SAR images in row i and column j are $M_1(i, j)$ and $M_2(i, j)$, respectively. The registration result is obtained by choosing the values of u' and v' that maximize the correlation coefficient R. Subsequently, the horizontal and vertical coordinates of the calibrated SAR data are matched with those of the uncalibrated SAR data one by one to reduce the error caused by image mismatch.

Given that this study uses the data blocks as the preparatory data for calculating the calibration constant, the presented technique in this study allows for a certain degree of tolerance toward registration errors [56].

In the registered target image, urban area and bare land scenes are segmented and divided into individual data blocks. Each data block measures 20 × 20 pixels, and each scene contains 20 data blocks. Subsequently, the previously discussed extraction method is employed to obtain DN-$\sigma^0_{\text{uncalSAR}}$ data block for calculating the calibration constant.

D. Correction Results of Incidence Angle Difference

In this section, we acquire four sets of historical image data from the cross-calibration satellites in the target scenes. These data aim to investigate the empirical relationship between the disparity in image amplitude values and the variation in backscattering coefficients. The objective is to establish a suitable threshold for scene evaluation. Detailed information regarding the datasets is provided in Table IV.

Two datasets from Sentinel-1A undergo preprocessing, excluding radiometric calibration, to obtain amplitude and backscattering data. The same operations are applied to the two sets of data from Sentinel-1B. Subsequently, the difference in backscattering coefficients and DN values for urban and bare land areas are calculated. A linear regression analysis is conducted to establish the empirical relationship between the two parameters. The specific findings are illustrated in Fig. 8.

Based on the findings depicted in Fig. 8, it is evident that in urban areas, a difference of 0.3 dB in backscattering coefficients $\Delta\sigma^0$ corresponds to a difference of 0.81 dB in amplitude values ΔDN. Thus, a threshold of 0.81 dB can be employed to evaluate the necessity of incidence angle correction in urban scene data. Equally, in the case of bare land scenes, a difference of 0.3 dB in backscattering coefficients $\Delta\sigma^0$ corresponds to a difference of 1.1 dB in amplitude values ΔDN. Consequently, a threshold of 1.1 dB can be utilized to

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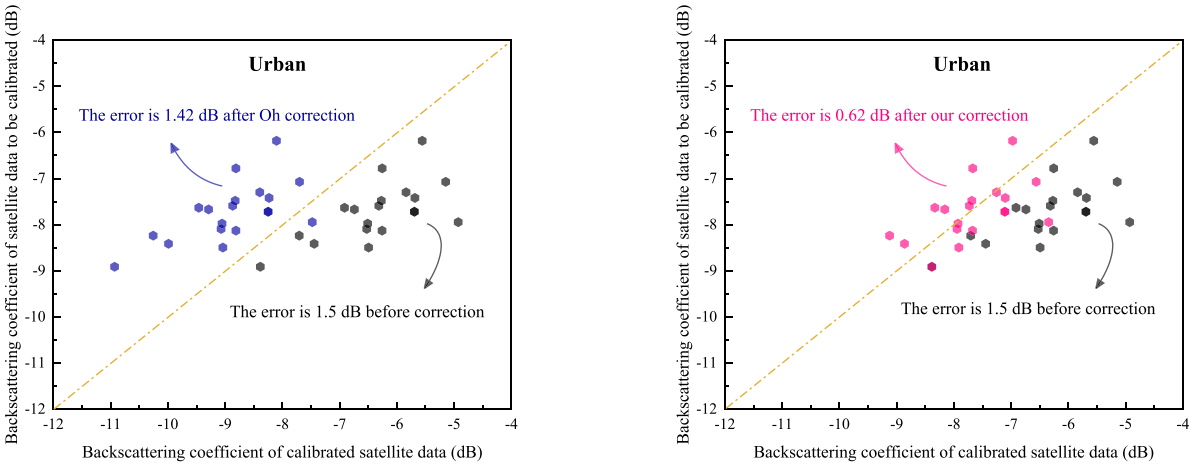


Fig. 9. Comparison of urban region incidence angle correction errors. (a) Comparison of the errors before and after incidence angle correction in the Oh model. (b) Comparison of the errors before and after incidence angle correction in the proposed method.

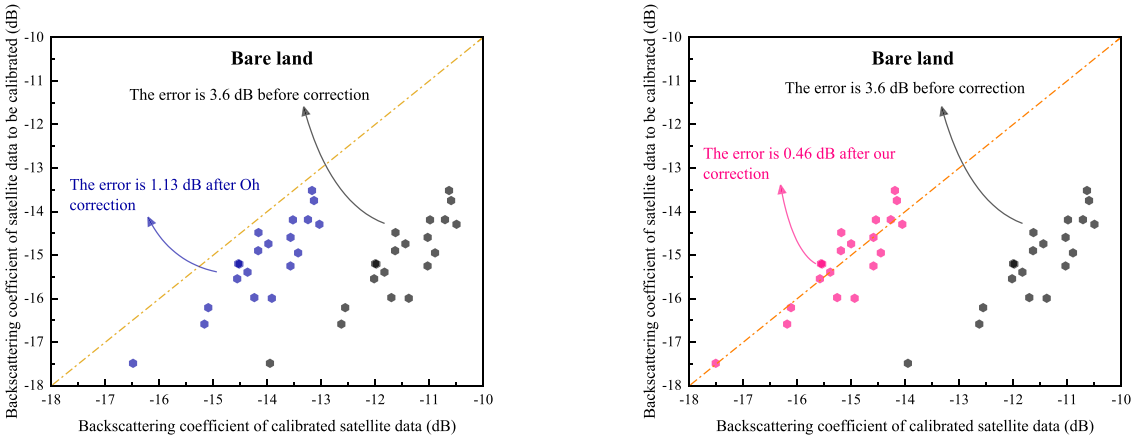


Fig. 10. Comparison of bare land region incidence angle correction errors. (a) Comparison of the errors before and after incidence angle correction in the Oh model. (b) Comparison of the errors before and after incidence angle correction in the proposed method.

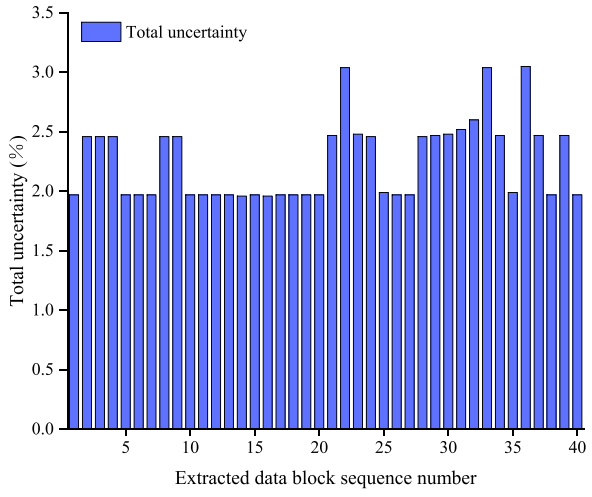


Fig. 11. Total uncertainty per data block.

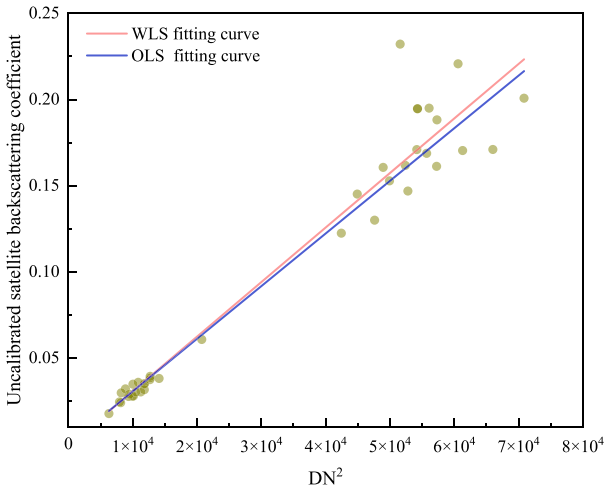
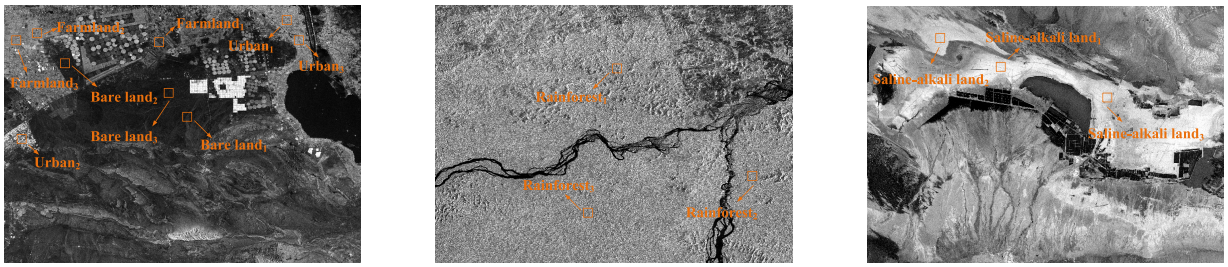


Fig. 12. Regression calibration lines for WLS and OLS.

determine the requirement for incidence angle correction in bare land scene data.

After applying the threshold values to filter the data blocks, the remaining data blocks are processed for incidence angle correction using the cosine model. The performance of this method is subsequently compared with the conventional approach utilizing the Oh model. The comparative analysis is presented in Figs. 9 and 10.

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(a) (b) (c)

Fig. 13. Specific position of the selected verification area. (a) Other areas in the same orbit at similar times, including urban, bare land, and farmland scenes. (b) Rainforest scenes from similar time periods. (c) Saline–alkali land scenes consistent with traditional method datasets.

It can be seen that the proposed method significantly reduces error in urban areas from Fig. 9. The error has effectively decreased from 1.5 to 0.62 dB. Compared with the Oh model-based method, the correction performance is enhanced by 0.8 dB. Similarly, Fig. 10 showcases the favorable outcomes of the proposed method in bare land, resulting in a remarkable reduction of error from 3.6 to 0.46 dB. Comparatively, the correction performance is improved by 0.67 dB compared with the Oh model-based method. In addition, our observations indicate that the Oh model-based method exhibits limited correction capability within urban area, indirectly implying its constrained correction range.

Although the correction method presented in this article is effective, the scene evaluation process entails the determination of both the distributed data block extraction method and the data block screening threshold. This process necessitates the utilization of time-series image data as well as historical image data. Consequently, the amount of preprocessing required escalates, and the scene evaluation leads to an increase in computational effort.

E. Calculation Results of Calibration Constant

In this section, we calculate the uncertainty of each data block to obtain the weight coefficient matrix. The total uncertainty for each data block is presented in Fig. 11, revealing a consistent value across most blocks. This uniformity results from carefully selecting a calibration scene that offers stability and uniformity. Consequently, data blocks captured under the same scenarios exhibit similar scattering stability and uniformity traits. Furthermore, the historical data for each block are acquired by the Sentinel-1 satellite under consistent imaging conditions, ensuring a relatively high level of spatial coherence. By combining the WLS method with the radiation calibration formula, we perform regression analysis on the calibration constants in urban area and bare land using 40 sets of data blocks.

For comparison purposes, we also employ the OLS method to regress the calibration constants for both urban area and bare land. The regression curves for both WLS and OLS approaches are depicted in Fig. 12. The extracted calibration formula is

$$\sigma^0_{\mathrm{WLS}} = 3.165 \times 10^{-6} \cdot \mathrm{DN}^2 - 0.0009599 \quad (27)$$

$$\sigma^0_{\mathrm{OLS}} = 3.053 \times 10^{-6} \cdot \mathrm{DN}^2 + 0.0000555 \quad (28)$$

F. Verification of Final Calibration Accuracy

To compare the calibration errors of the extracted calibration constants obtained from the WLS and OLS methods, we utilize these constants to verify additional scene areas that share the same time and orbital conditions. In selecting verification regions, we prioritized the following criteria to ensure their representativeness: 1) each verification region is a homogeneous area that captures the typical scattering characteristics inherent to its respective scene type, thereby ensuring accurate representation and 2) the verification regions encompass a diverse range of scene types to ensure the broad applicability of our findings. By adhering to these considerations, we have ensured the representativeness of the selected verification regions. Specifically, urban area, bare land, and farmland are selected as the scene regions for verification. To further demonstrate the advantages of the method in this article, image data that are temporally consistent with the dataset of the calibration constants derived from the traditional Oh model-based calibration are selected for verification. In addition, verification experiments are also conducted in the rainforest areas to evaluate the calibration constants' suitability.

In this article, three subregions in different scenarios, denoted by subscripts 1/2/3, are selected to evaluate the effectiveness of the SAR radiometric cross-calibration approach in this article. The specific locations of the verification regions selected for this article are shown in Fig. 13.

Based on the data presented in Table V, it can be seen that in all four verification scenes, the calibration errors for the WLS method are lower than those for the OLS method. The downward arrows in the table indicate this reduction in error, and the values to the right of them are the relative percentage of error reduction, signifying that the calibration performance of WLS is superior to that of OLS. The calibration constant derived from WLS exhibits an average reduction in the average relative error of approximately 21.6%, with a maximum decrease of 52.7%. Similarly, the average reduction in the RMSE of the calibration achieved by WLS is approximately 18.6%, with a maximum decrease of 51.6%. When examining the higher average relative error in urban$_2$ area, several factors come into play. Notably, this area exhibits a considerable presence of strong scattering points, as demonstrated by the calculated maximum backscattering coefficient of 8.95 dB, which significantly exceeds the average

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TABLE V
CALIBRATION ACCURACY OF CALIBRATION CONSTANTS OBTAINED BY WLS AND OLS IN DIFFERENT SCENARIOS

Scene	Region	OLS		WLS	
		ε	RMSE	ε	RMSE
Urban	Urban_1	0.1168	0.6132	**0.0952 ↓ 18.5%**	**0.5000 ↓ 18.5%**
	Urban_2	0.4627	0.5685	**0.3326 ↓ 28.1%**	**0.4557 ↓ 19.8%**
	Urban_3	0.4321	0.6661	**0.3516 ↓ 18.6%**	**0.5790 ↓ 13.1%**
Bare land	Bare land_1	0.0866	0.7102	**0.0705 ↓ 18.6%**	**0.6050 ↓ 14.8%**
	Bare land_2	0.0670	0.7739	**0.0551 ↓ 17.8%**	**0.5780 ↓ 25.3%**
	Bare land_2	0.0124	0.2045	**0.0102 ↓ 17.7%**	**0.1690 ↓ 17.4%**
Farmland	Farmland_1	0.0061	0.1660	**0.0042 ↓ 31.1%**	**0.1091 ↓ 34.2%**
	Farmland_2	0.0264	0.5560	**0.0125 ↓ 52.7%**	**0.2689 ↓ 51.6%**
	Farmland_3	0.0397	0.4848	**0.0311 ↓ 21.7%**	**0.4034 ↓ 16.8%**
Rainforest	Rainforest_1	0.0635	0.5469	**0.0480 ↓ 24.4%**	**0.4381 ↓ 19.9%**
	Rainforest_2	0.0429	0.4112	**0.0295 ↓ 31.2%**	**0.3333 ↓ 18.9%**
	Rainforest_2	0.1344	1.0318	**0.1165 ↓ 13.3%**	**0.9028 ↓ 12.5%**
Saline-alkali land	$\text{Saline-alkali land}_1$	0.0611	0.6366	**0.0511 ↓ 16.3%**	**0.5583 ↓ 12.2%**
	$\text{Saline-alkali land}_2$	0.0585	0.6159	**0.0491 ↓ 13.3%**	**0.5438 ↓ 11.7%**
	$\text{Saline-alkali land}_2$	0.0541	0.6249	**0.0461 ↓ 14.8%**	**0.5639 ↓ 9.8%**
Mean		0.1109	0.5740	**0.0869 ↓ 21.6%**	**0.4672 ↓ 18.6%**

backscattering coefficient for the region. Consequently, this discrepancy contributes to the more significant relative error observed. Furthermore, this discrepancy indirectly suggests a diminished level of uniformity within the urban_2 area.

V. CONCLUSION

This article proposes an improved SAR radiometric cross-calibration method based on scene-driven incidence angle difference correction, which addresses the limitations of the traditional radiometric cross-calibration method regarding applicable scenes and overcorrection of data blocks. In addition, this method incorporates weighting coefficients for each regression data block to account for quality uncertainty differences among the data blocks. The calibrated satellite utilized in this study is Sentinel-1A, which applies incidence angle correction and regression calibration constant to the uncalibrated Sentinel-1B. The results indicate that the incidence angle correction method proposed in this article exhibits superior performance in urban areas and bare land compared with the traditional Oh model. The correction error is reduced by 0.8 and 0.67 dB, respectively. Moreover, using the WLS method for regression analysis of calibration constant yields a decrease in the ε by approximately 21.6% and a reduction in the RMSE by about 18.6% compared with the OLS method. Despite these advancements, the method is currently experimented with and validated only on data from bare land and urban areas, leaving some limitations that may hinder its applicability in more complex or heterogeneous environments. In addition, the method has been tested exclusively on Sentinel-1 series satellite data, with no exploration of its effectiveness on other satellite platforms. Future research will aim to apply this method to a broader range of data types to assess its efficacy across various scene types. Furthermore, investigations will extend to the applicability of the method on different satellite platforms beyond Sentinel-1, while also examining the influence of additional sources of uncertainty on the calibration constant.

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REFERENCES

[1] A. Moreira, P. Prats-Iraola, M. Younis, G. Krieger, I. Hajnsek, and K. P. Papathanassiou, "A tutorial on synthetic aperture radar," *IEEE Geosci. Remote Sens. Mag.*, vol. 1, no. 1, pp. 6–43, Mar. 2013.

[2] Y. Deng, W. Yu, H. Zhang, W. Wang, D. Liu, and R. Wang, "Forthcoming spaceborne SAR development," *J. Radars*, vol. 9, no. 1, pp. 1–33, Feb. 2020.

[3] F. Ma, F. Zhang, Q. Yin, D. Xiang, and Y. Zhou, "Fast SAR image segmentation with deep task-specific superpixel sampling and soft graph convolution," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5214116.

[4] F. Ma, F. Zhang, D. Xiang, Q. Yin, and Y. Zhou, "Fast task-specific region merging for SAR image segmentation," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5222316.

[5] F. Ma, X. Sun, F. Zhang, Y. Zhou, and H.-C. Li, "What catch your attention in SAR images: Saliency detection based on soft-superpixel lacunarity cue," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5200817.

[6] Y. Zhou, H. Liu, F. Ma, Z. Pan, and F. Zhang, "A sidelobe-aware small ship detection network for synthetic aperture radar imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5205516.

[7] Z. Yue et al., "A novel semi-supervised convolutional neural network method for synthetic aperture radar image recognition," *Cogn. Comput.*, vol. 13, no. 4, pp. 795–806, Jul. 2021.

[8] A. Freeman, "SAR calibration: An overview," *IEEE Trans. Geosci. Remote Sens.*, vol. 30, no. 6, pp. 1107–1121, Nov. 1992.

[9] T. Lepekhina and V. Nikolaev, "Spaceborne SAR radiometric calibration and characterization using a radar target simulator," in *Proc. Russian Open Conf. Radio Wave Propag. (RWP)*, vol. 1, Jul. 2019, pp. 333–336.

[10] P. Imperatore, R. Lanari, and A. Pepe, "Gical: Geo-morphometric inverse cylindrical method for radiometric calibration of SAR images," in *Proc. IEEE Int. Geosci. Remote Sens. Symp. (IGARSS)*, Jul. 2018, pp. 7836–7839.

[11] H. Li, H. Zhang, Q. Chen, Y. Gao, X. Shi, and L. Zhang, "Accuracy analysis of radiometric calibration in-orbit for SuperView Neo-2 SAR satellite," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 17, pp. 2234–2244, 2024.

[12] F. Gao et al., "SAR target incremental recognition based on features with strong separability," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5202813.

[13] H. Huang, F. Gao, J. Sun, J. Wang, A. Hussain, and H. Zhou, "Novel category discovery without forgetting for automatic target recognition," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 17, pp. 4408–4420, 2024.

[14] Y. Zhou, C. Li, L. Ma, M. Y. Yang, and Q. Liu, "Improved trihedral corner reflector for high-precision SAR calibration and validation," in *Proc. IEEE Geosci. Remote Sens. Symp.*, Jul. 2014, pp. 454–457.

[15] W. Liang, Z. Jia, J. Hong, Q. Zhang, A. Wang, and Z. Deng, "Polarimetric calibration scheme combining internal and external calibrations, and experiment for Gaofen-3," *IEEE Access*, vol. 8, pp. 7659–7671, 2020.

[16] A. Piccioni, R. Alesii, F. Santucci, and F. Graziosi, "Software-defined corner reflector for satellite SAR systems," in *Proc. IEEE Aerosp. Conf. (AERO)*, Mar. 2022, pp. 1–7.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 09:27:11 UTC from IEEE Xplore. Restrictions apply.

[17] P. Guccione, M. Scagliola, and D. Giudici, "Low-frequency SAR radiometric calibration and antenna pattern estimation by using stable point targets," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 2, pp. 635–646, Feb. 2018.

[18] J. Lu, T. He, S. Liang, and Y. Zhang, "An automatic radiometric cross-calibration method for wide-angle medium-resolution multispectral satellite sensor using Landsat data," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5604011.

[19] C. Niu et al., "Radiometric cross-calibration of the ZY1-02D hyperspectral imager using the GF-5 AHSI imager," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5519612.

[20] A. Yang, B. Zhong, S. Wu, and Q. Liu, "Radiometric cross-calibration of GF-4 in multispectral bands," *Remote Sens.*, vol. 9, no. 3, p. 232, Mar. 2017.

[21] Y. Jia, J. Yang, M. Lin, Y. Zhang, C. Ma, and C. Fan, "Global assessments of the HY-2B measurements and cross-calibrations with Jason-3," *Remote Sens.*, vol. 12, no. 15, p. 2470, Aug. 2020.

[22] J. Huang, B. Yan, and N. Sun, "Monitoring of the cross-calibration biases between the S-NPP and NOAA-20 VIIRS sensor data records using goes advanced baseline imager as a transfer," in *Proc. IEEE Int. Geosci. Remote Sens. Symp. (IGARSS)*, Sep. 2020, pp. 6393–6396.

[23] Y. Zhou, L. Zhuang, J. Duan, F. Zhang, and W. Hong, "Synthetic aperture radar radiometric cross calibration based on distributed targets," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 15, pp. 9599–9612, 2022.

[24] D. Xiang, W. Li, X. Sun, H. Wang, and Y. Su, "Sidelobe suppression for high-resolution SAR imagery based on spectral reshaping and feature statistical difference," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5211314.

[25] D. Xiang, X. Pan, H. Ding, J. Cheng, and X. Sun, "Two-stage registration of SAR images with large distortion based on superpixel segmentation," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5211115.

[26] F. Zhang, X. Yao, H. Tang, Q. Yin, Y. Hu, and B. Lei, "Multiple mode SAR raw data simulation and parallel acceleration for Gaofen-3 mission," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 11, no. 6, pp. 2115–2126, Jun. 2018.

[27] I. E. Mladenova, T. J. Jackson, R. Bindlish, and S. Hensley, "Incidence angle normalization of radar backscatter data," *IEEE Trans. Geosci. Remote Sens.*, vol. 51, no. 3, pp. 1791–1804, Mar. 2013.

[28] T. Geldsetzer and S. E. L. Howell, "Incidence angle dependencies for C-band backscatter from sea ice during both the winter and melt season," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 4302515.

[29] J. P. Ardila, V. Tolpekin, and W. Bijker, "Angular backscatter variation in L-band ALOS ScanSAR images of tropical forest areas," *IEEE Geosci. Remote Sens. Lett.*, vol. 7, no. 4, pp. 821–825, Oct. 2010.

[30] B. Yang, Y. Zhou, Q. Yin, F. Ma, and F. Zhang, "An effective approach to mitigate the influence of incidence angle difference on SAR radiometric cross-calibration through scene evaluation," in *Proc. SAR Big Data Era (BIGSARDATA)*, Sep. 2023, pp. 1–4.

[31] L. Li, F. Zhang, Y. Shao, Q. Wei, Q. Huang, and Y. Jiao, "Airborne SAR radiometric calibration based on improved sliding window integral method," *Sensors*, vol. 22, no. 1, p. 320, Jan. 2022.

[32] R. J. Carroll, C. F. J. Wu, and D. Ruppert, "The effect of estimating weights in weighted least squares," *J. Amer. Stat. Assoc.*, vol. 83, no. 404, pp. 1045–1054, 1988.

[33] M. Zink, P. Olivier, and A. Freeman, "Cross-calibration between airborne SAR sensors," *IEEE Trans. Geosci. Remote Sens.*, vol. 31, no. 1, pp. 237–245, Jun. 1993.

[34] H. Wakabayashi, T. Tadono, M. Matsuoka, T. Matsuoka, M. Shimada, and S. Uratsuka, "Cross-calibration experiment of airborne L-band polarimetric SAR," in *Proc. Scanning Present Resolving Future. IEEE Int. Geosci. Remote Sens. Symp.*, vol. 1, Jul. 2001, pp. 423–425.

[35] E. Uratsuka et al., "Combination of two airborne SAR observations (PacRim2 over Japan)," in *Proc. Scanning Present Resolving Future. IEEE Int. Geosci. Remote Sens. Symp.*, vol. 3, Sep. 2001, pp. 1400–1402.

[36] P. Z. Alexander Zakharov and A. Sokolov, "Intercalibration of ERS AMI and Envisat ASAR with ground-based," in *Proc. Envisat & ERS Symp.*, 2004, pp. 6–10.

[37] D. O'Grady, M. Leblanc, and D. Gillieson, "Relationship of local incidence angle with satellite radar backscatter for different surface conditions," *Int. J. Appl. Earth Observ. Geoinf.*, vol. 24, pp. 42–53, Oct. 2013.

[38] B. Widhalm, A. Bartsch, A. Roth, and M. Leibman, "Classification of Tundra regions with polarimetric Terrasar-X data," in *Proc. IGARSS - IEEE Int. Geosci. Remote Sens. Symp.*, Jul. 2018, pp. 8551–8554.

[39] X. Wu, J. P. Walker, N. N. Das, R. Panciera, and C. Rüdiger, "Evaluation of the SMAP brightness temperature downscaling algorithm using active–passive microwave observations," *Remote Sens. Environ.*, vol. 155, pp. 210–221, Dec. 2014.

[40] N. Ye, J. P. Walker, and C. Rüdiger, "A cumulative distribution function method for normalizing variable-angle microwave observations," *IEEE Trans. Geosci. Remote Sens.*, vol. 53, no. 7, pp. 3906–3916, Jul. 2015.

[41] W. Lang, P. Zhang, J. Wu, Y. Shen, and X. Yang, "Incidence angle correction of SAR sea ice data based on locally linear mapping," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 6, pp. 3188–3199, Jun. 2016.

[42] F. T. Ulaby, F. Kouyate, A. K. Fung, and A. J. Sieber, "A backscatter model for a randomly perturbed periodic surface," *IEEE Trans. Geosci. Remote Sens.*, vol. GRS-20, no. 4, pp. 518–528, Oct. 1982.

[43] P. Guo, T. Zhao, J. Shi, H. Xu, X. Li, and S. Niu, "Assessing the active-passive approach at variant incidence angles for microwave brightness temperature downscaling," *Int. J. Digit. Earth*, vol. 14, no. 10, pp. 1273–1293, Oct. 2021.

[44] E. F. Vermote and N. Z. Saleous, "Calibration of NOAA16 AVHRR over a desert site using MODIS data," *Remote Sens. Environ.*, vol. 105, no. 3, pp. 214–220, Dec. 2006.

[45] W. Zhao, X. Hu, W. Fang, Y. Wang, and N. Xu, "Development and applications of intercalibration for satellite optical instruments," *Opt. Precis. Eng.*, vol. 23, no. 7, pp. 1921–1931, 2015.

[46] Z. Chen et al., "On-orbit performance assessment of Beijing-1 microsatellite sensors," *J. Remote Sens.*, vol. 12, no. 3, pp. 468–476, 2008.

[47] S. Shangguan, X. Qiu, and K. Fu, "Research on a special hyper-pixel for SAR radiometric monitoring," *Remote Sens.*, vol. 15, no. 8, p. 2175, Apr. 2023.

[48] J. Yang, X. Qiu, C. Ding, and B. Lei, "Identification of stable backscattering features, suitable for maintaining absolute synthetic aperture radar (SAR) radiometric calibration of Sentinel-1," *Remote Sens.*, vol. 10, no. 7, p. 1010, Jun. 2018.

[49] X. Chen, Y. Zhou, F. Ma, Q. Yin, and F. Zhang, "Optimized selection of pseudo-invariant calibration sites for radiometric cross-calibration of spaceborne SAR sensors," in *Proc. SAR Big Data Era (BIGSARDATA)*, Sep. 2023, pp. 1–4.

[50] M. Skolnik, *Radar Handbook*. New York, NY, USA: McGraw-Hill, 1990.

[51] A. Freeman, "Radiometric calibration of SAR image data," *Int. Arch. Photogramm. Remote Sens.*, vol. 29, p. 212, Nov. 1993.

[52] Y. Deng, H. Zheng, R. Wang, J. Feng, and Y. Liu, "Internal calibration for stepped-frequency chirp SAR imaging," *IEEE Geosci. Remote Sens. Lett.*, vol. 8, no. 6, pp. 1105–1109, Nov. 2011.

[53] P. Wang, H. Sun, and W. Yu, "A novel wireless internal calibration method of spaceborne SAR," *J. Radars.*, vol. 7, no. 4, pp. 425–436, 2018.

[54] M. Schwerdt et al., "Independent verification of the Sentinel-1A system calibration," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 9, no. 3, pp. 994–1007, Mar. 2016.

[55] C. Li, Z. Yu, and J. Chen, "Overview of techniques for improving high-resolution spaceborne SAR imaging and image quality," *J. Radars*, vol. 8, no. 6, pp. 717–731, 2019.

[56] D. Xiang, H. Ding, X. Sun, J. Cheng, C. Hu, and Y. Su, "PolSAR image registration combining Siamese multiscale attention network and joint filter," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5208414.



Yongsheng Zhou (Member, IEEE) received the B.E. degree in communication engineering from Beijing Information Science and Technology University, Beijing, China, in 2005, and the Ph.D. degree in signal and information processing from the Institute of Electronics, Chinese Academy of Sciences, Beijing, in 2010.

He was with the Academy of Opto-Electronics, Chinese Academy of Sciences, from 2010 to 2019. He is currently a Professor of electronic and information engineering with the College of Information Science and Technology, Beijing University of Chemical Technology. His research interests include target detection and recognition from microwave remotely sensed images, digital signal, and image processing.

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Bopeng Yang (Student Member, IEEE) is pursuing the master's degree with the College of Information Science and Technology, Beijing University of Chemical Technology, Beijing, China.

He is majoring in New Generation Electronic Information Technology. His research interests include SAR image processing and SAR calibration.



Qiang Yin (Senior Member, IEEE) received the B.S. degree in electronic and information engineering from Beijing University of Chemical Technology, Beijing, China, in 2004, and the M.S. and Ph.D. degrees in signal and information processing from the Institute of Electronics, Chinese Academy of Science, Beijing, in 2008 and 2016, respectively.

From 2008 to 2013, she was a Research Assistant at the Institute of Electronics, Chinese Academy of Sciences. From 2014 to 2015, she was a Research Fellow at the European Space Agency, Rome, Italy. She is currently an Associate Professor with the College of Information Science and Technology, Beijing University of Chemical Technology. Her research interests include polarimetric/polarimetric interferometric synthetic aperture radar (SAR) and deep learning.



Fei Ma (Member, IEEE) received the B.S., M.S., and Ph.D. degrees in electronic and information engineering from Beijing University of Aeronautics and Astronautics (BUAA), Beijing, China, in 2013, 2016, and 2020, respectively.

He is currently an Associate Professor with the College of Information Science and Technology, Beijing University of Chemical Technology, Beijing. His research interests include radar signal processing, image processing, machine learning, and target detection.



Fan Zhang (Senior Member, IEEE) received the B.E. degree in communication engineering from the Civil Aviation University of China, Tianjin, China, in 2002, the M.S. degree in signal and information processing from Beihang University, Beijing, China, in 2005, and the Ph.D. degree in signal and information processing from the Institute of Electronics, Chinese Academy of Sciences, Beijing, in 2008.

From 2013 to 2015, he was a Visiting Scholar working with Prof. Martin D. F. Wong with the Coordinated Science Laboratory, University of Illinois Urbana-Champaign, Urbana, IL, USA. He is currently a Full Professor of electronic and information engineering with the Beijing University of Chemical Technology, Beijing. He has authored or coauthored more than 100 peerreviewed articles, including IEEE TRANSACTIONS ON GEOSCIENCE AND REMOTE SENSING, *ISPRS Journal of Photogrammetry and remote sensing*, CVPR, ECCV and ICCV. His research interests include synthetic aperture radar system simulation, SAR image interpretation, and artificial intelligence.

Dr. Zhang is an Editorial Board Member of *Journal of Radars*.

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